Omid Mokhtari

Ph.D. Student — Power Systems

Contact Information

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Research Interests

Network reduction for power systems; optimization; scalable computation (decomposition, parallel and GPU computing); voltage estimation and sensor placement; network science; power system planning and operation; electricity markets.

Education

University of Vermont Aug. 2023 – Present

Ph.D. in Electrical Engineering, Burlington, Vermont, USA

Supervisors: Dr. Samuel Chevalier and Dr. Mads R. Almassalkhi. Research focus: scalable optimization and network reduction methods for electric power networks. Expected completion: Fall 2027.

Sharif University of Technology Sept. 2017 – Jan. 2020

M.Sc. in Electrical Engineering – Power Systems, Tehran, Iran

Thesis: "Co-planning of Multiple Merchant Energy Storage Systems and Transmission Systems." Supervisor: Prof. Seyed Hamid Hosseini.

Imam Khomeini International University of Qazvin Sept. 2012 – July 2017

B.Sc. in Electrical Engineering – Power Systems, Qazvin, Iran

Final project: "Investigation of Wind Turbines Dynamic Behavior During a Fault, Using DIgSILENT." Supervisor: Dr. Navid Ghaffarzadeh.

Publications

Journal articles

[J1] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Structure-preserving optimal Kron-based reduction of radial distribution networks," *IEEE Transactions on Power Systems*, early access, 2026, doi: 10.1109/TPWRS.2026.3726028.

[J2] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Optimal Kron-based reduction of networks (Opti-KRON) for three-phase distribution feeders," *Electric Power Systems Research*, vol. 263, Art. no. 113615, 2027 (PSCC 2026 paper), doi: 10.1016/j.epsr.2026.113615.

Preprints

[P1] O. Molodchyk, **O. Mokhtari**, S. Chevalier, M. R. Almassalkhi, and T. Faulwasser, "Data-driven power flow for radial distribution networks with sparse real-time data," arXiv preprint arXiv:2604.08403, 2026.

[P2] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Enhancing scalability of optimal Kron-based reduction of networks (Opti-KRON) via decomposition with community detection," arXiv preprint arXiv:2407.02679, 2024.

Poster Presentations

[1] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Structure-preserving optimal Kron-based reduction of radial distribution networks," PowerUp, Boulder, CO, 2026.

[2] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Optimal Kron-based reduction of networks (Opti-KRON) for three-phase distribution feeders," Student Research Conference (SRC), University of Vermont, 2026.

[3] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Network reduction and radialization," Student Research Conference (SRC), University of Vermont, 2025.

[4] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Optimal network reduction," IEEE Power & Energy Society General Meeting (PESGM), Seattle, WA, 2024.

[5] **O. Mokhtari**, S. Chevalier, and M. R. Almassalkhi, "Community detection based network reduction," Student Research Conference (SRC), University of Vermont, 2024.

Professional Service

Reviewer for *IEEE Transactions on Power Systems*, *Electric Power Systems Research*, and the American Control Conference (ACC).

Technical Skills

Programming	Julia, Python, MATLAB, GAMS
Software	DIgSILENT PowerFactory, LaTeX